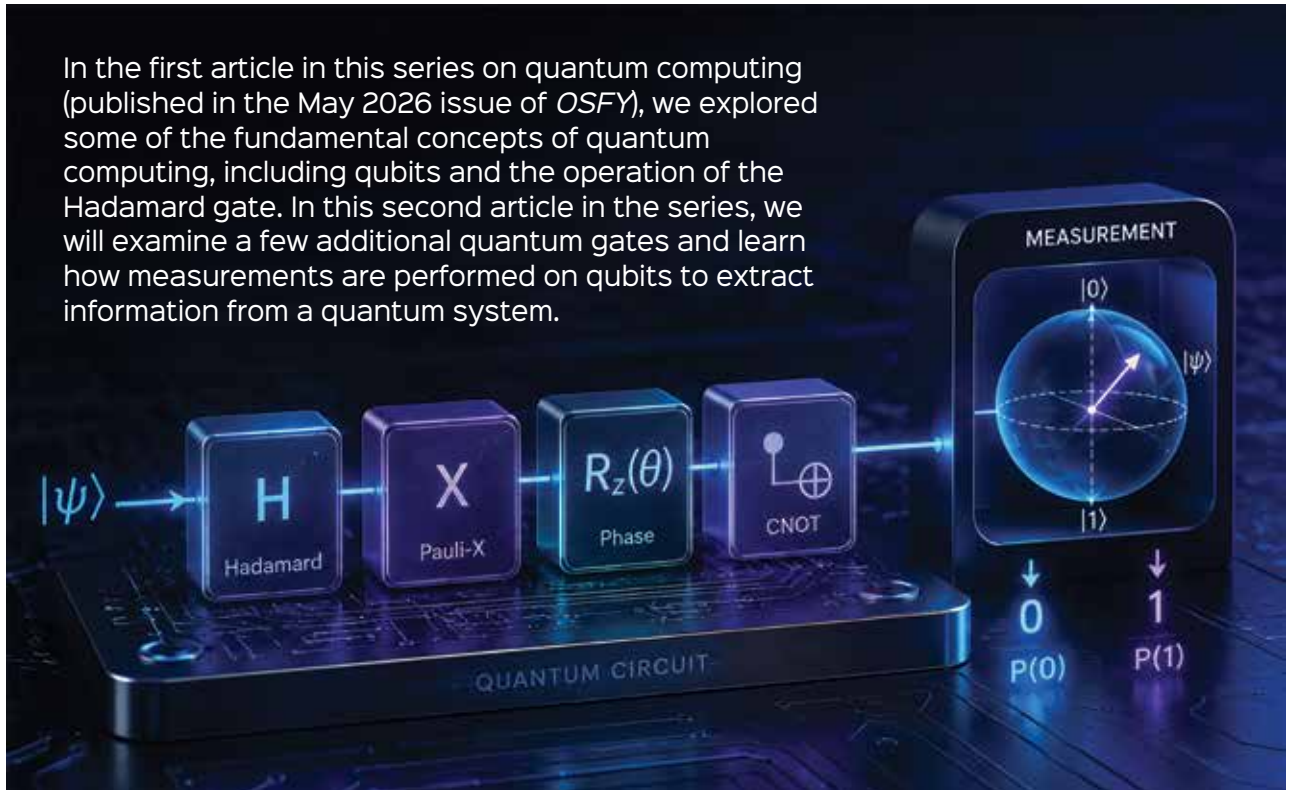


Quantum Computing: Gates and Measurement

In the first article in this series on quantum computing (published in the May 2026 issue of *OSFY*), we explored some of the fundamental concepts of quantum computing, including qubits and the operation of the Hadamard gate. In this second article in the series, we will examine a few additional quantum gates and learn how measurements are performed on qubits to extract information from a quantum system.



As discussed in the previous article, our exploration of quantum computing will take place at three levels: the coding level, the mathematical level, and the hardware level. While we plan to use several platforms and programming languages to develop quantum algorithms throughout this series, we will begin with Qiskit, IBM's quantum software development kit (SDK). In the previous article, we installed Qiskit and successfully executed our first quantum program.

Another theme we will explore throughout this series is the history of quantum computing and the latest developments in the field. In the

previous article, we discussed the motivation behind quantum computers and the two groundbreaking algorithms that paved the way for practical quantum computing: Shor's algorithm and Grover's algorithm. In this article, we turn our attention to the various hardware architectures that have been proposed, tested, and developed in the quest to build a practical quantum computer.

Today, there are several competing approaches to building quantum computers, and it is still unclear which architecture will ultimately lead to large scale, fault-tolerant quantum computers. Some of the leading approaches include superconducting qubits, trapped-ion

quantum computers, neutral-atom quantum computers, photonic quantum computers, etc. At present, no one can say with certainty which of these architectures will eventually dominate.

The situation bears some resemblance to the early days of classical computing. In the late 1940s and early 1950s, nobody knew whether the newly invented transistor would eventually replace vacuum tubes. Early transistors were expensive, relatively immature, and offered limited performance, while vacuum tubes had decades of engineering experience behind them and formed the basis of virtually all electronic computers of the time. Nevertheless, continuous